# DoangJoo "Alan" Synn

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## Research Interests

I study **motion** as a computational medium in computer graphics and human-computer interaction (HCI). I build interactive systems for designing, generating, and fabricating motion, connecting expressive intent with computational design and physical realization. My research interests include computer animation, computational design, motion synthesis, and AI-assisted creative tools. My earlier work spans machine-learning systems, cloud software, games, and social robots.

## Education

**Georgia Institute of Technology** Aug. 2022-Present
*Ph.D. Candidate in Computer Science*

- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

**Korea University** Mar. 2020-Feb. 2022
*M.Eng. in Electrical and Computer Engineering*

- Thesis: *On Efficient Data Delivery for Machine Learning Systems*
- Advised by Prof. Jong-Kook Kim

**Korea University** Mar. 2012-Feb. 2020
*B.Eng. in Electrical Engineering*

- Alternative military service as a Software Engineer, 2016-2019

## Selected Research Projects

**MotionSmith** CHI 2026
*Sketch-based design of mechanical motion and fabricable artifacts*

- Developed a computational design system that connects sketched motion paths, mechanism generation, interactive refinement, and fabrication-ready output.
- Developed through a year-long participatory design process with three expert automata makers; supported motion exploration and the fabrication of working mechanical artifacts.
- Publication C1; project page and source code.

**MotionSmith for Young Makers** Manuscript
*Educator co-design and classroom enactment of a mechanical CAD system*

- Extended the research to classroom use through co-design with four K-8 STEM educators, revising the authoring workflow, reusable fabrication kit, and instructional supports.
- Used in educator-led classroom activities with 129 middle-school students at two U.S. schools; three educators facilitated STEM, STEAM, and engineering activities over three to four sessions per class.
- Manuscript M1.

## Publications

### Refereed Conference Papers

**C1** **DoangJoo Synn**, Zhifan Guo, Sehoon Ha, and HyunJoo Oh. "MotionSmith: A Sketch-Based Design System for Automata Making." *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. doi:10.1145/3772318.3791545.

**C2** Jin Yu, Poojita Garg, **DoangJoo Synn**, and HyunJoo Oh. "Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning." *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025. doi:10.1145/3706598.3713260.

**C3** **DoangJoo Synn** and Jong-Kook Kim. "Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders." *Proceedings of ACK 2021*, 2021. doi:10.3745/PKIPS.y2021m11a.446.

**C4** Byung Hoon Ahn, **DoangJoo Synn**, Masih Derkani, Eiman Ebrahimi, and Hadi Esmaeilzadeh. "Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection." *Neural Information Processing Systems Demonstrations*, 2021.

**Refereed Journal Articles**

**J1** **DoangJoo Synn**, XinYu Piao, JooYoung Park, and Jong-Kook Kim. "Distributed Training of Deep Learning Models Using Micro Batch Streaming." *KIPS Transactions on Software and Data Engineering*, 2023. doi:10.3745/KTSDE.2023.12.3.101.

## Patent Applications

**IP1** HyunJoo Oh, Jin Yu, **DoangJoo Synn**, and Poojita Garg. "Coding Blocks Learning System and Method." Published U.S. patent application US 2026/0112292 A1, 23 April 2026. Application No. 19/357,594; filed 14 October 2025. Applicant: Georgia Tech Research Corporation.

## Professional Experience

**Dolby Laboratories** May 2024-Aug. 2025
*Ph.D. Research Intern, Experience Delivery Lab*

- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

**LG Uplus** June-Aug. 2023
*MLOps Engineering Intern*

- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

**Protopia AI** Jan. 2021-Aug. 2022
*Research Scientist*

- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.
- Led framework development and Kubernetes/GPU infrastructure integration; the work contributed to a NeurIPS 2021 Demo.

**Cloud Architect · AWS Educate Cloud Ambassador** 2020-2022

- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.
- Built cloud-native infrastructure using Kubernetes, Terraform, CI/CD, observability, and cost-optimized autoscaling while mentoring early-career engineers.

**Flysher** Jan.-Dec. 2019
*Server Engineer*

- Shipped backend features for four social-game titles; one reached Facebook's global top 10.
- Built an ETL-to-Redshift analytics pipeline supporting data-informed feature rollout.

**AKA Intelligence** Apr. 2017-Jan. 2019
*System Engineer*

- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.
- Coordinated hardware-software integration and production QA with overseas ODMs.

## Teaching

**Georgia Institute of Technology** Aug. 2024-Present
*Head Teaching Assistant, CS 4496/7496: Computer Animation*

- School of Interactive Computing.

**Korea University Science & Technology School** Feb. 2021; Jan. 2022
*Lecturer and Program Developer*

- Taught introductory Java programming to 10+ undergraduate students; contributed to the BK21 Socialware IT education program.

**Korea University** Spring; Fall 2021
*Teaching Assistant*

- Digital System Design
- Data Structures and Algorithms

**KAIST Engineering School** Jan. 2012-Aug. 2014
*Lecturer and Curriculum Developer*

- Co-developed national project-based coding education programs with South Korea's Ministry of Science and ICT.
- Designed and evaluated 20+ curricula combining Arduino, hardware prototyping, and software development for high-school and college students.

## Invited Talks & Outreach

**Invited Panelist and MotionSmith Demonstrator, AutomataCon** 2026
**Motion, Making & Mechanical Design: GIFT Workshop, Georgia Tech** 2026
**CEISMC**
**Invited Talk, Korea University** 2026
**MotionSmith Presentation, CHI** 2026

## Scholarly Activities

**Reviewer, CHI** 2027
**Reviewer, CHI** 2026
**Subreviewer, CHI** 2025
**Artifact Evaluation Committee, ASPLOS 2024** 2024

## Honors & Awards

**Fellowships**

- Brain Korea 21 (BK21) Research Fellowship (2020-2022)

**Awards**

- Prize Winner, Creative Challenge Program @ Korea Univ. CTL (2015 and 2016)

## Activities & Service

**Georgia Tech Korean Student Association** 2023-2024

- President of Korean Students in the College of Computing.

**Hardware & Software Community, Korea University** 2014-2019

- President; grew the community from 10 to 200+ active members across hardware and software tracks.

**TEDxKoreaUniversity, Korea University** 2015-2016

- Founder & Chief Organizer; led a 47-person team in producing a 200-guest event with 7 speakers and a $4K budget.